

Large Language Models: How They Work

What Is a Large Language Model?

An LLM is a massively scaled-up version of smartphone predictive text. Your phone's keyboard predicts the next word based on a few recent words. An LLM predicts the next token based on billions of parameters trained on trillions of pages of text — books, Wikipedia, code, web pages.

Core mechanism: LLMs are next-token predictors. Everything else — reasoning, summarising, coding — emerges from predicting text extremely well at scale.

Training an LLM — The Three Phases

Phase 1: Pre-training: Feed the model trillions of text tokens. Objective: predict the next token.
Duration: weeks to months on thousands of GPUs.

Phase 2: Supervised Fine-Tuning (SFT): Train on high-quality human-written demonstrations of desired behaviour (helpful answers, safe refusals).

Phase 3: RLHF: Reinforcement Learning from Human Feedback. Human raters rank model outputs; a reward model learns their preferences; the LLM is fine-tuned to maximise reward.

RLHF is what transforms a raw text predictor into a helpful, harmless, honest assistant.

Emergent Capabilities

As LLMs scale, unexpected abilities emerge that were not explicitly trained:

- In-context learning (learning from a few examples in the prompt).
- Chain-of-thought reasoning (solving multi-step problems step by step).
- Code generation across dozens of programming languages.

LLM Limitations to Know

Hallucination: Confident generation of false facts. The model predicts plausible text, not verified truth.

Knowledge cutoff: Training data has a fixed end date. The model is 'frozen in time'.

Context window: Limited working memory — older context gets 'forgotten' in very long conversations.

No persistent memory: By default, each conversation starts fresh with no memory of past interactions.

Key Takeaways

- LLMs are next-token predictors scaled to billions of parameters on trillions of tokens.
- Training: pre-training → SFT → RLHF.
- Emergent capabilities arise from scale: in-context learning, reasoning, code generation.
- Key limits: hallucination, knowledge cutoff, finite context window.

Quick-Revision Questions

- Explain LLMs using the smartphone keyboard analogy.
- What is RLHF and why is it needed?
- Name three limitations of current LLMs.